

Zhenmiao Zhang

Postdoctoral Scholar in Computer Science, University of California, San Diego

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Research Interests

- **Telomere-to-Telomere (T2T) Genome Assembly:** Designing computational algorithms to resolve long-standing challenges in achieving gapless genome reconstructions. My work integrates classical graph theory (e.g., de Bruijn and string graphs) with machine learning to resolve complex genomic regions, aiming to reconstruct complete genomes with maximum efficiency and accuracy.
- **Metagenomics & Microbial Dark Matter:** Developing computational frameworks to assemble and bin high-quality genomes from complex environmental samples. By leveraging self-supervised learning and sequence analysis, my research aims to systematically recover “microbial dark matter”—previously unculturable or unobserved species—to push the boundaries of “Complete Metagenomics.”
- **Comparative Genomics & Evolution:** Building bioinformatics software to extract biological insights from the emerging era of T2T genomes. My work focuses on the automated identification of large-scale genomic features, such as segmental duplications and syntenic blocks, to characterize structural variation and reconstruct the evolutionary trajectories that drive biological diversity.

Education & Academic Appointments

Postdoctoral Scholar-Employee, Department of Computer Science and Engineering 05/2024 – Present

University of California, San Diego (UCSD), USA

Advisor: Prof. Pavel A. Pevzner

Doctor of Philosophy, Department of Computer Science

09/2019 – 07/2024

Hong Kong Baptist University, Hong Kong, China

Advisor: Prof. Lu Zhang

Thesis: *Advanced Computational Methods to Decipher Microbial Dark Matter Using Metagenomic Sequencing.*

Visiting Graduate Student, Department of Computer Science and Engineering 02/2023 – 08/2023

University of California, San Diego (UCSD), USA

Advisor: Prof. Pavel A. Pevzner

Visiting Graduate Student, Department of Computer Science and Engineering 11/2023 – 02/2024

University of California, San Diego (UCSD), USA

Advisor: Prof. Pavel A. Pevzner

B.Eng. in Computer Science & B.Sc. in Mathematics

09/2015 – 06/2019

Renmin University of China (RUC), China

Publications (Selected, * means correspondence)

Bohao Zou, **Zhenmiao Zhang (co-first)**, Xiaohan Wang, Rong Tao, Nianzhen Gu, Karsten Kristiansen, Mo Han, Lu Zhang*. *CompleteBin: A transformer-based framework unlocks microbial dark matter through improved short contig binning.* Preparing for resubmission to **Nature Biotechnology (IF=41.7)**.

Zhenmiao Zhang*, Marcus W. Fedarko, Anton Bankevich, Pavel A. Pevzner. *Haplotype-mixed assembly of long and accurate reads.* **Genome Biology (IF=9.4)**, 2026.

Jingjing Wang, **Zhenmiao Zhang (co-first)**, Yang Chen, Xi Zhou, Jiaxin Xiang, Chao Yang, Dmitry A Rodionov, Andrei L Osterman, Qinwei Qiu, Yusheng Deng, Yanmin Liu, Chengrui Wang, Xiaoxiao Shang, Li Huang, Chen Sun, Jianwen Guo, Zhimin Yang, Lijuan Han, Lixiang Zhai, Zhaoxiang Bian, Wei Jia*, Xiaodong Fang*, Lu Zhang*. *Exploring Population Differences in the Human Gut Microbiome: from*

Microbial Abundance to Single Nucleotide Polymorphisms. **Genome Biology** (IF=9.4), 2026.

Zhenmiao Zhang*, Ishaan Gupta, Pavel A. Pevzner. *GenomeDecoder: Inferring Segmental Duplications in Highly-Repetitive Genomic Regions*. **Bioinformatics** (IF=5.4), 2025.

T2T Consortium; **Zhenmiao Zhang** (contributing author). *Complete sequencing of ape genomes*. **Nature** (IF=48.5), 2025.

Zhenmiao Zhang, Jin Xiao, Hongbo Wang, Chao Yang, Yufen Huang, Zhen Yue, Yang Chen, Lijuan Han, Kejing Yin, Aiping Lyu, Xiaodong Fang, Lu Zhang*. *Exploring high-quality microbial genomes by assembling short reads with long-range connectivity*. **Nature Communications** (IF=15.7), 2024.

Chao Yang, **Zhenmiao Zhang (co-first)**, Yufen Huang, Xuefeng Xie, Herui Liao, Jin Xiao, Werner Pieter Veldsman, Kejing Yin, Xiaodong Fang, Lu Zhang*. *LRTK: a platform-agnostic toolkit for linked-read analysis of both human genome and metagenome*. **GigaScience** (IF=3.9), 2024.

Zhenmiao Zhang, Chao Yang, Werner Pieter Veldsman, Xiaodong Fang, Lu Zhang*. *Benchmarking genome assembly methods on metagenomic sequencing data*. **Briefings in Bioinformatics** (IF=7.7), 2023.

Zhenmiao Zhang, Lu Zhang*. *METAMVGL: a multi-view graph-based metagenomic contig binning algorithm by integrating assembly and paired-end graphs*. **BMC Bioinformatics** (IF=3.3), 2021.

Conferences

- *Haplotype-mixed assembly of long and accurate reads*, PAG 2026 (Oral and Poster)
- *Improve metagenomic binning by recovering short contigs using deep learning*, RECOMB 2023 (Poster).
- *Exploring high-quality microbial genomes by assembly of linked-reads with high barcode specificity using deep learning*, RECOMB-SEQ 2023 (Oral).
- *METAMVGL: a multi-view graph-based metagenomic contig binning algorithm by integrating assembly and PE graphs*, APBC 2021 (Oral).

Software & Algorithmic Contributions

- **Mosaic Genome Assembler** — De novo haplotype-mixed genome assembler for HiFi reads
- **GenomeDecoder** — Framework for inferring syntenic blocks in complete genomes
- **CompleteBin** — Contig binning via dynamic contrastive learning with pretrained language models
- **Pangaea** — Metagenome assembly with long-range connectivity using deep variational autoencoder
- **METAMVGL** — Contig binning using multi-view graph learning integrating assembly and PE graphs

Teaching Experience

- COMP7250 Machine Learning
- COMP1007 Introduction to Python (2020-2023)

Academic Service

- **Reviewer**: *Nature Communications*, *RECOMB*, *Microbiome*, *Gut Microbes*, *NAR Genomics and Bioinformatics*, *Scientific Reports*, *Computational Biology and Chemistry*, *BMC Methods*, *BIBM*, and others
- **PC Member**: BIBM 2026, BIBM 2025, BIBM 2025 workshops on *Computational methods to characterize genomic variants using high-throughput sequencing data*, BIBM 2022 workshops on *Algorithmic Advances for Single-Cell and Spatial Omics Data Analysis*

Grants & Honors

- UCSD CSE Fellows Program (Postdoctoral Funding)
- Open Project Grant (BGI-Shenzhen): *Novel algorithms of metagenome assembly and haplotype phasing using single-tube Long Fragment Read sequencing technology*
- Excellent Teaching Assistant Performance Award (COMP1007 Python), 2020–2023